

Skill - 6

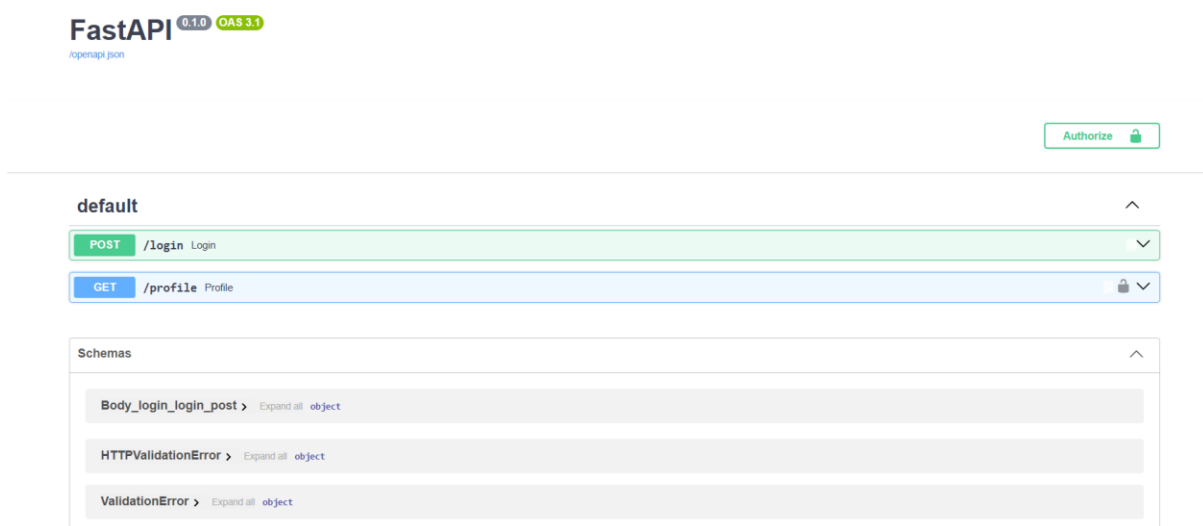
The screenshot shows a VS Code editor with a project named 'Skill6DB'. The Explorer sidebar on the left shows the file structure with a 'main.py' file. The main editor window displays the code for 'main.py', which is a FastAPI application. The code includes imports for FastAPI, HTTPException, Depends, OAuth2PasswordBearer, OAuth2PasswordRequestForm, and jwt. It defines a FastAPI app, sets a secret key and algorithm, and creates an OAuth2 password bearer scheme. A login endpoint is defined with a POST method, and a profile endpoint is defined with a GET method. The login endpoint checks the username and password and returns a JWT token if the credentials are correct. The profile endpoint returns the user's profile information. The bottom panel shows the terminal output, which includes the command to install python-multipart and the command to run the application with uvicorn. The terminal output shows that the application is running on http://127.0.0.1:8000.

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15
16     # Classroom example only
17     if form_data.username == "101" and form_data.password == "abc123":
18
19         payload = {
20             "sub": form_data.username,
21             "role": "student"
22         }
23
```

Terminal Output:

```
PS C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB> python -m pip install python-multipart
Requirement already satisfied: python-multipart in C:\Users\birad\AppData\Local\Programs\Python\Python314\Lib\site-packages (0.0.32)
PS C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [2812] using StatReload
INFO: Started server process [3900]
```

Swagger:



POST /Login:

default

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

```
grant_type
string | (string |
null)
pattern: ^password$

username * required
string

password * required
string($password)

scope
string

client_id
string | (string |
null)

client_secret
string | (string |
null)($password)
```

Authorize in Swagger:

astAPI 0.1.0 OAS 3.1

default

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

grant_type string | (string | null) pattern: ^password\$

username * required string

password * required string(\$password)

scope string

☒ Send empty value

Available authorizations

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes. API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Token URL: login

Flow: password

username:

password:

Client credentials location: Authorization header

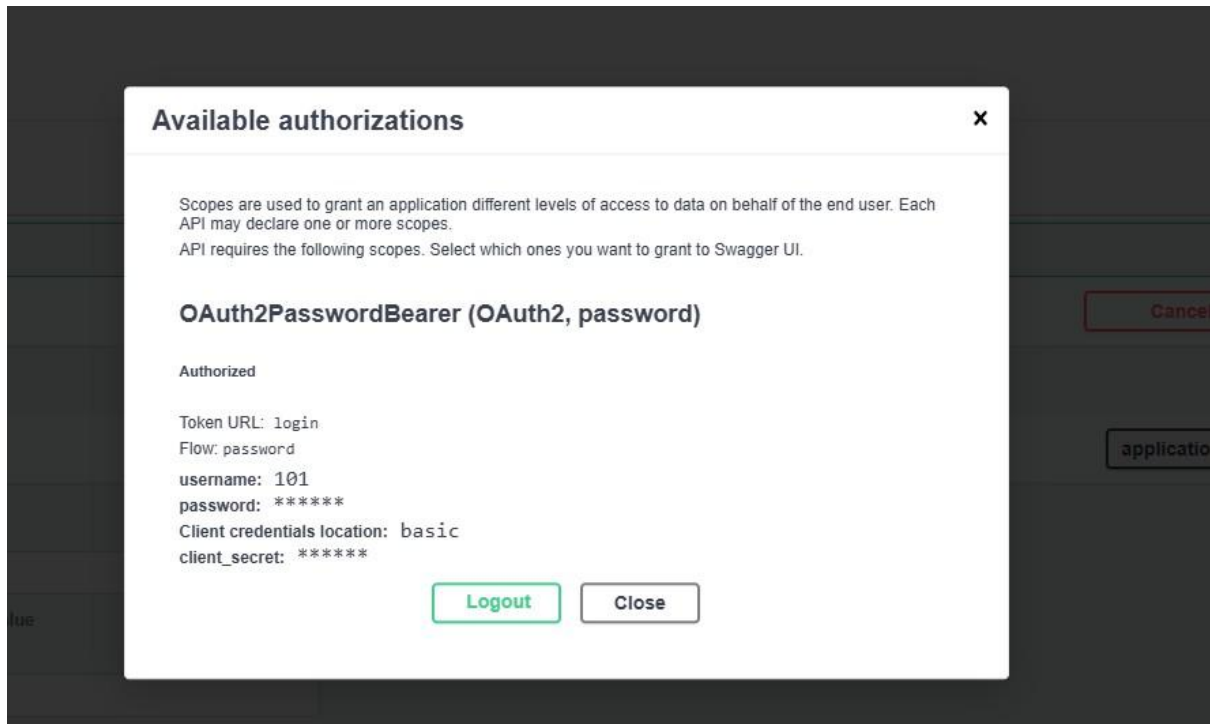
client_id:

client_secret:

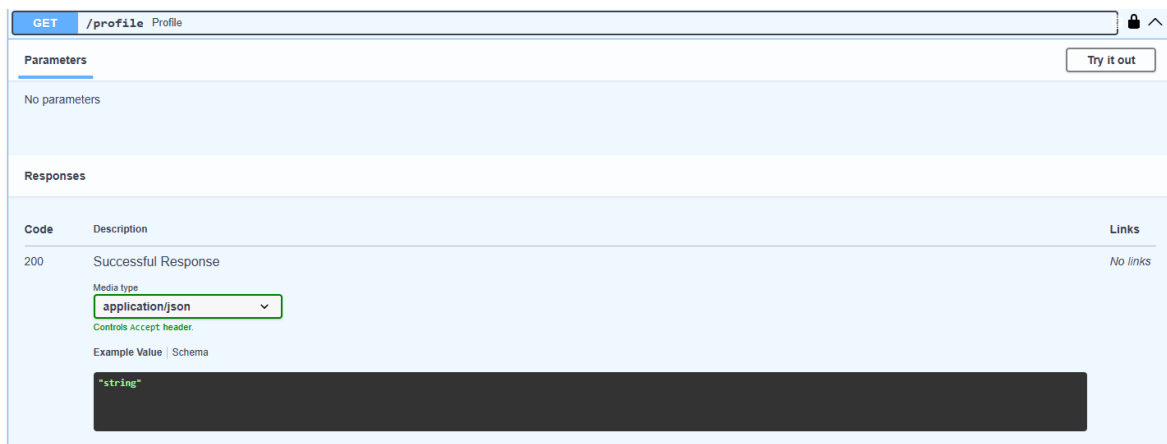
Authorize Close

Authorize

Cancel Reset



GET/Profile:



GET

/profile Profile

Parameters

No parameters

Cancel

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDEiLCJyb2xiIjoic3RlZGVudCJ9.NhkwZDlPBjIqSDFQ1kndoWt0jblUWkHw4uRQ6PK1zI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to your profile", "user_id": "101", "role": "student" }</pre> Download Response headers <pre>content-length: 70 content-type: application/json date: Fri, 18 Sep 2026 04:07:00 GMT server: uvicorn</pre>

Responses

Code	Description	Links
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